

Generative AI Healthcare Agents

Team 3:

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Data Collection

Key Data Sources

MIMIC-III, IV (PhysioNet)

- 380K+ ICU patients, 2M+ clinical notes
- 6 CSV files: Diagnosis, EDStays, MedRecon, Pyxis, Triage, Vitalsign
- Rows: 7,195,726
- Includes: Vitals, diagnoses, meds, triage scores, demographics
- URL: physionet.org

iCliniq (Hugging Face)

- Online telemedicine consultations
- 39,540 records
- Fields: Title, Question, Answer

MedQuAD (U.S. NLM)

- Expert-verified medical Q&A
- 16,412 records
- Fields: Question, Answer, Source, Focus_area

Storage: All datasets uploaded to **Google Cloud Platform**



Google Cloud



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Search



Composer / Environments

Environments



CREATE



REFRESH



DELETE



Filter Filter environments

☐	State	Name ↑	Location	Composer version	Airflow version	Creation time	Update time	Airflow webserver	DAG list	Logs	DAGs folder	Labels
☐	✓	project298	us-central1	3	2.10.2-build.12	4/9/25, 1:06 AM	4/9/25, 4:14 AM	Airflow	DAGs	Logs	DAGs	None

Data Ingestion

- **Process** - Uploaded CSV files to GCP Cloud Storage bucket gs://data298a.
- **Security** - Configured IAM roles to restrict access to project team and pipeline services.
- **Features** - Enabled versioning to track data history, Logged metadata (e.g., upload timestamps) for traceability.

Cloud Composer Setup -

- Environment: healthcare-composer (3 nodes, 2 vCPUs, 4 GB memory each).
- Installed PyPI packages: pandas, google-cloud-storage, apache-airflow[gcp].
- Auto-generated GKE cluster and gs://healthcare-composer-bucket/dags.

Centralized storage with ~7M rows, ready for transformation

Google Cloud

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Search

Cloud Storage

Bucket details

Go to path

Refresh

Learn

Overview

Buckets

Monitoring

Settings

Storage Intelligence

Insights datasets

Configuration

Marketplace

Release Notes

Objects

Configuration

Permissions

Protection

Lifecycle

Observability

Inventory Reports

Operations

Folder browser

Buckets

>

data298a

Create folder

Upload

Transfer data

Other services

Filter by name prefix only

Filter

Filter objects and folders

Show

Live objects only

	Name	Size	Type	Created	Storage class	Location
☐	diagnosis.csv	52.2 MB	text/csv	Apr 9, 2025, 12:54:49 AM	Standard	A
☐	edstays.csv	39.5 MB	text/csv	Apr 9, 2025, 12:54:47 AM	Standard	A
☐	final/	—	Folder	—	—	—
☐	icd_codes.csv	14.5 MB	text/csv	Apr 9, 2025, 1:19:00 AM	Standard	A
☐	intermediate/	—	Folder	—	—	—
☐	medrecon.csv	376.9 MB	text/csv	Apr 9, 2025, 12:54:51 AM	Standard	A
☐	processed/	—	Folder	—	—	—
☐	pyxis.csv	110.4 MB	text/csv	Apr 9, 2025, 12:54:45 AM	Standard	A
☐	triage.csv	38.7 MB	text/csv	Apr 9, 2025, 12:54:47 AM	Standard	A
☐	vitalsign.csv	120.3 MB	text/csv	Apr 9, 2025, 12:54:45 AM	Standard	A

Data Preprocessing

- **Cleaning:** Removed null values, special characters, excessive whitespace
- **Normalization:** Lowercase text, standardized formats
- **Stop Word Removal:** Removed common words (e.g., *the*, *is*, *at*) to reduce noise
- **Stemming & Lemmatization:** Reduced words to their root/base form for semantic consistency
- **Field Merging:** Combined *title + question* (iCliniq), merged multi-source tables (MIMIC)
- **ICD Code Mapping:** Mapped ICD-9/10 codes to clinical descriptions for interpretability
- **Pre-tokenization & Embedding:** Vectorized text fields using BioBERT
- **LLM-Ready Outputs:** Final JSONL formatted for fine-tuning and Retrieval-Augmented Generation (RAG)

Clear task

Mark state as...

Filter DAG by task

Details Graph Gantt <> Code Event Log **Logs** XCom K8s Pod Spec Task Duration

All Levels

All File Sources

☐ Wrap

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```
[2025-04-09, 19:25:40 UTC] {dag2.py:76} INFO - Processing chunk with shape: (50000, 29)
[2025-04-09, 19:25:46 UTC] {dag2.py:90} INFO - Number of unique ICD codes in chunk: 1448
[2025-04-09, 19:25:46 UTC] {dag2.py:91} INFO - Top 10 ICD code frequencies in chunk:\nacd_code\n4019      2056\nI10      1738\nn25000      1455\nnE119      1006\nnR109      923\nnR079      916\nnZ794
[2025-04-09, 19:26:26 UTC] {dag2.py:99} INFO - Processed and appended chunk to processed/merged_mimic_IV.csv
[2025-04-09, 19:26:26 UTC] {dag2.py:76} INFO - Processing chunk with shape: (50000, 29)
[2025-04-09, 19:26:26 UTC] {dag2.py:90} INFO - Number of unique ICD codes in chunk: 1433
[2025-04-09, 19:26:26 UTC] {dag2.py:91} INFO - Top 10 ICD code frequencies in chunk:\nacd_code\n4019      1949\nI10      1398\nn25000      1308\nnR109      801\nnZ794      712\nn78650      667\nnE119
[2025-04-09, 19:26:55 UTC] {dag2.py:99} INFO - Processed and appended chunk to processed/merged_mimic_IV.csv
[2025-04-09, 19:26:56 UTC] {dag2.py:76} INFO - Processing chunk with shape: (50000, 29)
[2025-04-09, 19:26:56 UTC] {dag2.py:90} INFO - Number of unique ICD codes in chunk: 1488
[2025-04-09, 19:26:56 UTC] {dag2.py:91} INFO - Top 10 ICD code frequencies in chunk:\nacd_code\n4019      2112\nI10      1430\nnR109      743\nnR0600      725\nn25000      699\nn78650      656\nnR079
[2025-04-09, 19:27:26 UTC] {dag2.py:99} INFO - Processed and appended chunk to processed/merged_mimic_IV.csv
[2025-04-09, 19:27:27 UTC] {dag2.py:76} INFO - Processing chunk with shape: (50000, 29)
[2025-04-09, 19:27:27 UTC] {dag2.py:90} INFO - Number of unique ICD codes in chunk: 1579
[2025-04-09, 19:27:27 UTC] {dag2.py:91} INFO - Top 10 ICD code frequencies in chunk:\nacd_code\n4019      2308\nI10      1558\nn25000      1426\nn5990      1039\nnE119      897\nn78650      709\nn2720
[2025-04-09, 19:27:55 UTC] {dag2.py:99} INFO - Processed and appended chunk to processed/merged_mimic_IV.csv
[2025-04-09, 19:27:55 UTC] {dag2.py:76} INFO - Processing chunk with shape: (50000, 29)
[2025-04-09, 19:27:55 UTC] {dag2.py:90} INFO - Number of unique ICD codes in chunk: 1614
[2025-04-09, 19:27:55 UTC] {dag2.py:91} INFO - Top 10 ICD code frequencies in chunk:\nacd_code\n4019      1758\nI10      1477\nn25000      1278\nnR079      621\nnE119      585\nn5990      577\nn4280
[2025-04-09, 19:28:26 UTC] {dag2.py:99} INFO - Processed and appended chunk to processed/merged_mimic_IV.csv
[2025-04-09, 19:28:27 UTC] {dag2.py:76} INFO - Processing chunk with shape: (50000, 29)
[2025-04-09, 19:28:27 UTC] {dag2.py:90} INFO - Number of unique ICD codes in chunk: 1509
[2025-04-09, 19:28:27 UTC] {dag2.py:91} INFO - Top 10 ICD code frequencies in chunk:\nacd_code\n4019      1824\nI10      1615\nn25000      1138\nnR079      1102\nnE119      793\nnR109      742\nnN390
[2025-04-09, 19:28:56 UTC] {dag2.py:99} INFO - Processed and appended chunk to processed/merged_mimic_IV.csv
[2025-04-09, 19:28:57 UTC] {dag2.py:76} INFO - Processing chunk with shape: (45726, 29)
[2025-04-09, 19:28:57 UTC] {dag2.py:90} INFO - Number of unique ICD codes in chunk: 1480
[2025-04-09, 19:28:57 UTC] {dag2.py:91} INFO - Top 10 ICD code frequencies in chunk:\nacd_code\n4019      1869\nI10      1388\nn25000      1007\nnR079      889\nn78650      810\nnE119      802\nnR0600
[2025-04-09, 19:29:29 UTC] {dag2.py:99} INFO - Processed and appended chunk to processed/merged_mimic_IV.csv
[2025-04-09, 19:29:29 UTC] {dag2.py:101} INFO - Merged data saved successfully!
[2025-04-09, 19:29:29 UTC] {python.py:240} INFO - Done. Returned value was: None
[2025-04-09, 19:29:29 UTC] {taskinstance.py:340} ▶ Post task execution logs
```

Data Transformation

- **Schema Flattening:** Merged MIMIC tables (edstays, diagnosis, vitals, meds, triage) into a single patient-stay record.
- **Feature Engineering:**
 - Extracted ICD categories & chapters from raw ICD-9/10 codes.
 - Integrated diagnostic labels with clinical descriptions.
 - Computed aggregated vitals (mean heart rate, BP, etc.) per patient stay.
- **Field Alignment:** Standardized columns (instruction, input, output) across MedQuAD, iCliniq, and MIMIC QA datasets for consistency.
- **Semantic Enrichment:** Augmented records with ICD descriptions, chief complaints, medications, and demographics.
- **Format Conversion:** Exported into **JSONL** files for LLM training and RAG pipelines.
- **Embedding Preparation:** Generated text embeddings (PubMed-BERT/BioBERT) stored in Pinecone for retrieval.

DAG

Run

Task

mimic_icd_processing_pipeline_low_memory / ▶ 2025-04-10, 18:34:10 UTC / process_icd_data

Clear task

Mark state as...

Filter DAG by task ▼

Details

Graph

Gantt

<> Code

Event Log

Logs

XCom

K8s Pod Spec

Task Duration

All Levels ▼

All File Sources ▼

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```
[2025-04-10, 18:38:09 UTC] {dag3.py:275} INFO - [18:38:09 UTC] Merge completed, chunk shape: (10000, 22)
[2025-04-10, 18:38:10 UTC] {dag3.py:279} INFO - [18:38:10 UTC] Post-processing completed for chunk 718
[2025-04-10, 18:38:10 UTC] {dag3.py:282} INFO - Grouped by ICD Category:\n                unique_admissions\nicd_category                \n038                1\n079
[2025-04-10, 18:38:10 UTC] {dag3.py:287} INFO - Quality Checks - Missing chapters: 17, Empty subject_id: 0, Missing icd_code: 17
[2025-04-10, 18:38:10 UTC] {dag3.py:291} INFO - [18:38:10 UTC] Writing chunk 718 to /tmp/full_output.csv, first_chunk=False
[2025-04-10, 18:38:10 UTC] {dag3.py:299} INFO - [18:38:10 UTC] Moving to next chunk, chunk_counter: 718
[2025-04-10, 18:38:10 UTC] {dag3.py:273} INFO - [18:38:10 UTC] Processing input chunk 719 with shape: (10000, 17)
[2025-04-10, 18:38:10 UTC] {dag3.py:275} INFO - [18:38:10 UTC] Merge completed, chunk shape: (10000, 22)
[2025-04-10, 18:38:10 UTC] {dag3.py:279} INFO - [18:38:10 UTC] Post-processing completed for chunk 719
[2025-04-10, 18:38:10 UTC] {dag3.py:282} INFO - Grouped by ICD Category:\n                unique_admissions\nicd_category                \n008                1\n034
[2025-04-10, 18:38:10 UTC] {dag3.py:287} INFO - Quality Checks - Missing chapters: 19, Empty subject_id: 0, Missing icd_code: 19
[2025-04-10, 18:38:10 UTC] {dag3.py:291} INFO - [18:38:10 UTC] Writing chunk 719 to /tmp/full_output.csv, first_chunk=False
[2025-04-10, 18:38:10 UTC] {dag3.py:299} INFO - [18:38:10 UTC] Moving to next chunk, chunk_counter: 719
[2025-04-10, 18:38:10 UTC] {dag3.py:273} INFO - [18:38:10 UTC] Processing input chunk 720 with shape: (5583, 17)
[2025-04-10, 18:38:10 UTC] {dag3.py:275} INFO - [18:38:10 UTC] Merge completed, chunk shape: (5583, 22)
[2025-04-10, 18:38:10 UTC] {dag3.py:279} INFO - [18:38:10 UTC] Post-processing completed for chunk 720
[2025-04-10, 18:38:10 UTC] {dag3.py:282} INFO - Grouped by ICD Category:\n                unique_admissions\nicd_category                \n079                0\n135
[2025-04-10, 18:38:10 UTC] {dag3.py:287} INFO - Quality Checks - Missing chapters: 1, Empty subject_id: 0, Missing icd_code: 1
[2025-04-10, 18:38:10 UTC] {dag3.py:291} INFO - [18:38:10 UTC] Writing chunk 720 to /tmp/full_output.csv, first_chunk=False
[2025-04-10, 18:38:10 UTC] {dag3.py:299} INFO - [18:38:10 UTC] Moving to next chunk, chunk_counter: 720
[2025-04-10, 18:38:10 UTC] {dag3.py:302} INFO - [18:38:10 UTC] All chunks processed, uploading to GCS: processed/final_processed_mimic.csv
[2025-04-10, 18:38:57 UTC] {dag3.py:50} INFO - [18:38:57 UTC] Writing DataFrame to temp file: /tmp/temp_output.csv, mode=w
[2025-04-10, 18:41:03 UTC] {dag3.py:55} INFO - [18:41:03 UTC] Temp file created, size: 1982653062 bytes
[2025-04-10, 18:41:03 UTC] {dag3.py:57} INFO - [18:41:03 UTC] Starting GCS upload to processed/final_processed_mimic.csv, first_chunk=True
[2025-04-10, 18:41:03 UTC] {dag3.py:62} INFO - [18:41:03 UTC] Uploading file to processed/final_processed_mimic.csv
[2025-04-10, 18:41:20 UTC] {dag3.py:64} INFO - [18:41:20 UTC] GCS upload completed successfully
[2025-04-10, 18:41:21 UTC] {dag3.py:79} INFO - [18:41:21 UTC] Cleaned up /tmp/temp_output.csv
[2025-04-10, 18:41:21 UTC] {dag3.py:304} INFO - DAG-3 completed successfully! Processed 720 chunks, total rows: 7195583
```

Data Preparation

- **Unified Dataset Creation:** Integrated QA datasets (MIMIC-III, IV, MedQuAD, iCliniq) into a single corpus.
- **Training Format Conversion:** Exported into **JSONL format** with fields:
 - *context* → background (e.g., title, clinical notes)
 - *input* → user query/question
 - *output* → ground-truth answer
- **Data Splitting:** Applied stratified split:
 - 80% → Training (fine-tuning LLMs)
 - 10% → Validation (hyperparameter tuning, early stopping)
 - 10% → Test (final unbiased evaluation)
- **Balance & Consistency Checks:** Verified schema alignment, ensured no data leakage, checked distribution of diagnoses, query lengths, and response types across splits.
- **Hosting & Scalability:** Stored datasets in **Google BigQuery** for large-scale queries and integration with GCP pipelines.

Google Cloud

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Bucket details

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Overview

Buckets

Monitoring

Settings

Storage Intelligence

Insights datasets

Configuration

Marketplace

Release Notes

ready-dataset

Locationus-east1 (South Carolina)Storage classStandardPublic accessNot publicProtectionSoft Delete

Objects

ConfigurationPermissionsProtectionLifecycleObservabilityNewInventory Reports

Folder browser

ready-dataset

test/

train/

val/

Buckets > ready-dataset

Create folderUploadTransfer dataOther services

Filter by name prefix onlyFilterFilter objects and foldersShowLive objects only

☐	Name	Size	Type	Created ?	Storage
☐	test/	—	Folder	—	—
☐	train/	—	Folder	—	—
☐	val/	—	Folder	—	—

Data Pipeline – Architecture & Workflow

Overview of DAGs (Directed Acyclic Graphs) - The pipeline consists of 5 key DAGs, each performing specific tasks to ingest, preprocess, and enrich data from the MIMIC III / IV dataset. These DAGs are orchestrated using **Apache Airflow** via **Cloud Composer** on GCP.

DAG 1:
mimic_merge_pipeline

Function: Merges raw data files into a unified dataset.

DAG 2:
mimic_preprocessing_pipeline

Function: Cleans and processes merged dataset for further analysis.

DAG 3:
mimic_icd_processing_pipeline_low_memory

Function: Performs ICD code enrichment and EDA.

DAG 4:
mimic_load_to_bigquery

Function: Loads the final data into bigquery

DAG 5:
master_mimic_pipeline

Function: Triggers all the other dags in sequence then perform log monitoring and validation



Airflow

DAGs

Cluster Activity

Datasets

Browse

Admin

Docs

Composer



17:33 UTC



project298

All 6

Active 6

Paused 0

Running 1

Failed 0

Filter DAGs by tag

Search DAGs



Auto-refresh



DAG	Owner	Runs	Schedule	Last Run	Next Run	Recent Tasks	Actions
airflow_monitoring	airflow		*10 * * * *	2025-04-10, 17:20:00	2025-04-10, 17:30:00		
master_mimic_pipeline mimic monthly orchestration scheduled	airflow		0 4 1 * *	2025-04-10, 16:36:03	2025-04-01, 04:00:00		
mimic_icd_processing_pipeline_low_memory gcs icd_processing low_memory mimic	airflow		None	2025-04-10, 17:22:12			
mimic_load_to_bigquery bigquery gcs mimic	airflow		None	2025-04-10, 17:29:17			
mimic_merge_pipeline gcs merge mimic	airflow		@daily	2025-04-10, 16:36:06	2025-04-10, 00:00:00		
mimic_preprocessing_pipeline gcs mimic preprocessing	airflow		@daily	2025-04-10, 16:41:09	2025-04-10, 00:00:00		

DAG: master_mimic_pipeline

Master DAG to orchestrate the MIMIC-III data pipeline

Schedule: 0 4 1 * *

Next Run ID: 2025-04-01, 04:00:00 UTC



04/10/2025

05:47:01 PM

All Run Types

All Run States

Clear Filters

Auto-refresh

25

Press **shift** + **/** for Shortcuts

deferred

failed

queued

removed

restarting

running

scheduled

shutdown

skipped

success

up_for_reschedule

up_for_retry

upstream_failed

no_status

DAG
master_mimic_pipeline / Run
2025-03-01, 04:00:00 UTC

Clear

Mark state as...

Details Graph Gantt Code Event Log

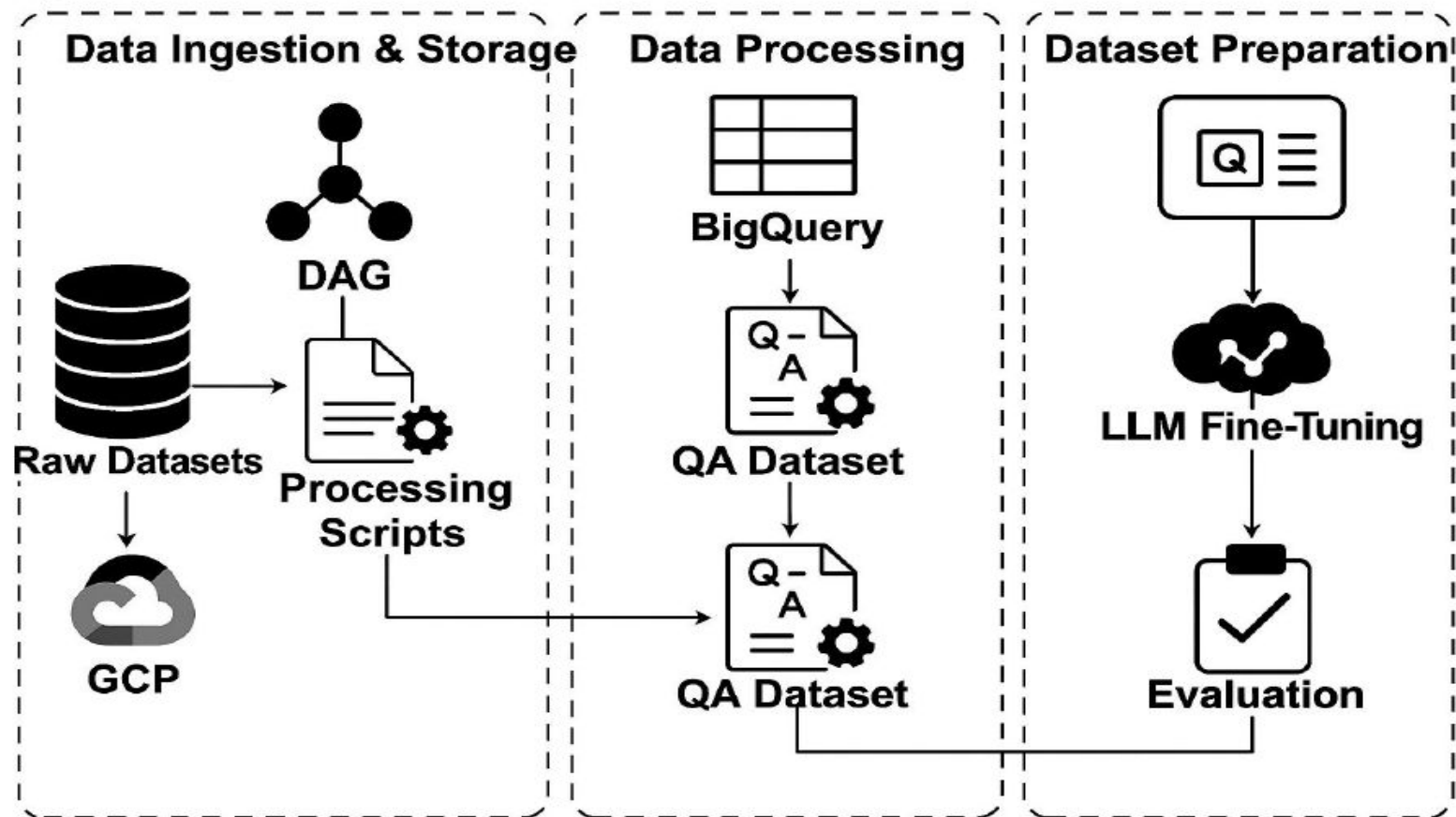
Layout:

Left -> Right

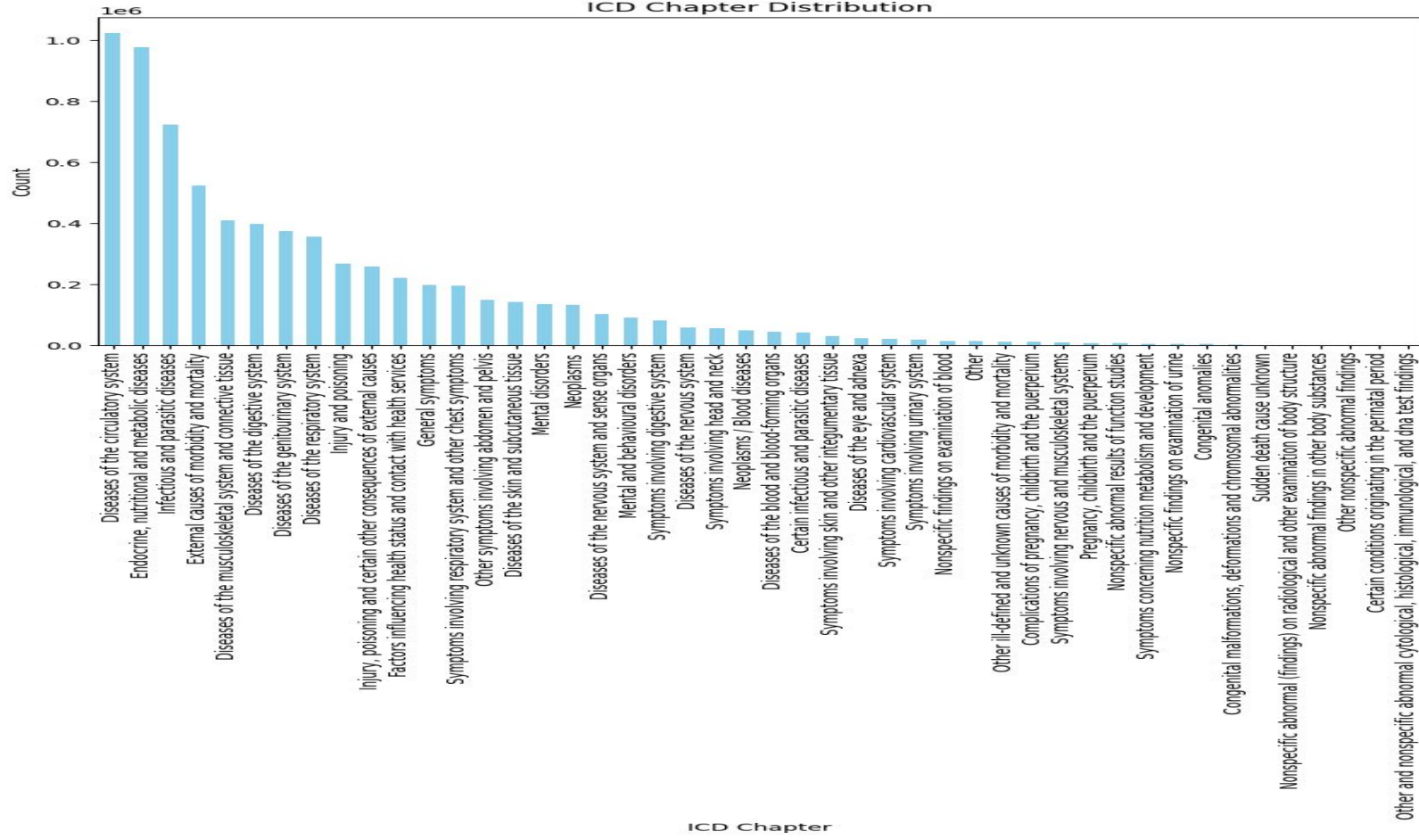
Task	Duration	State
start_pipeline	01:07:27	success
trigger_mimic_merge	00:33:43	success
trigger_mimic_preprocess	00:00:00	success
trigger_mimic_lcd_process		success
trigger_load_to_bigquery		success
validate_bigquery_table		success
monitor_pipeline_logs		success
end_pipeline		success

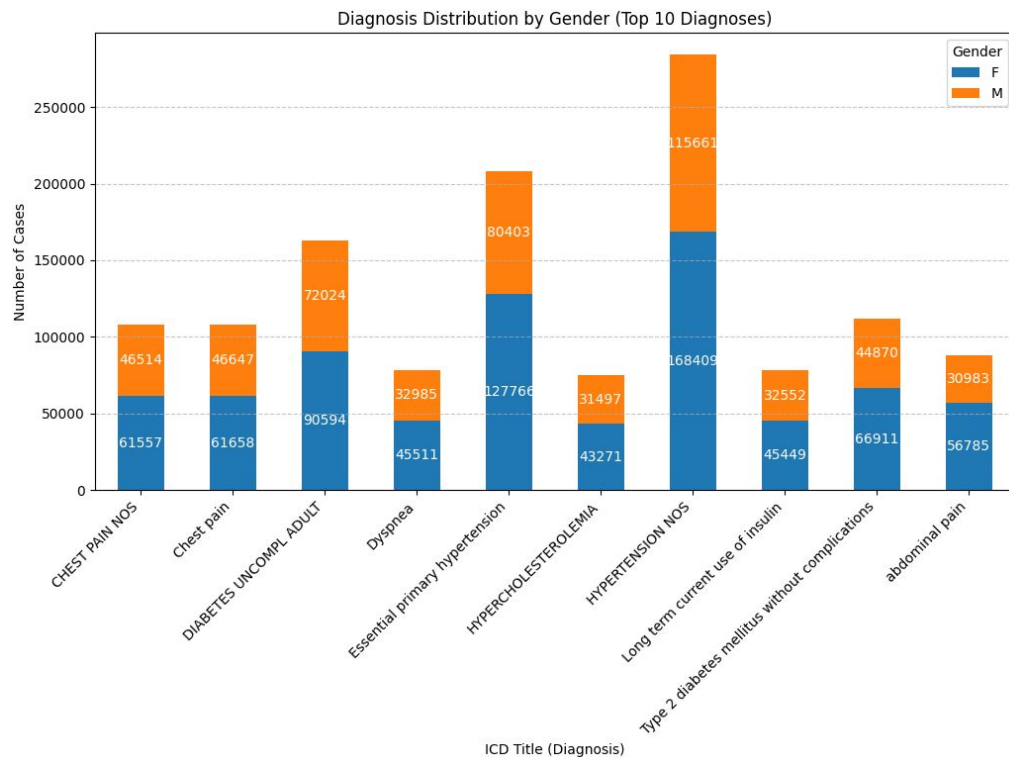
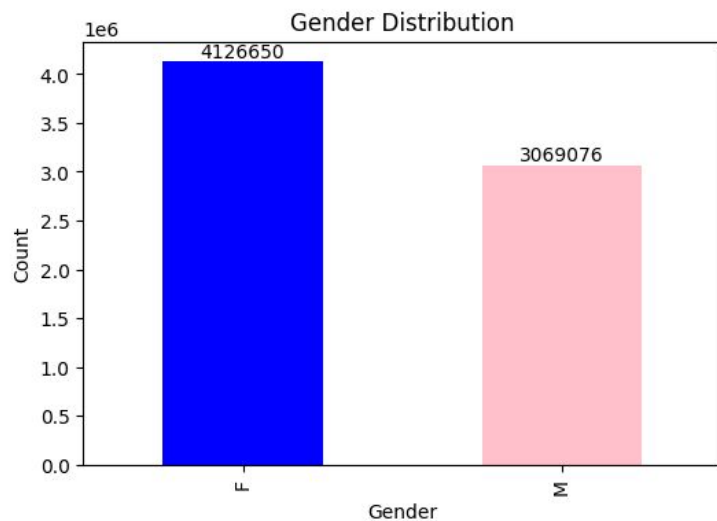


Figure: End-to-End Architecture

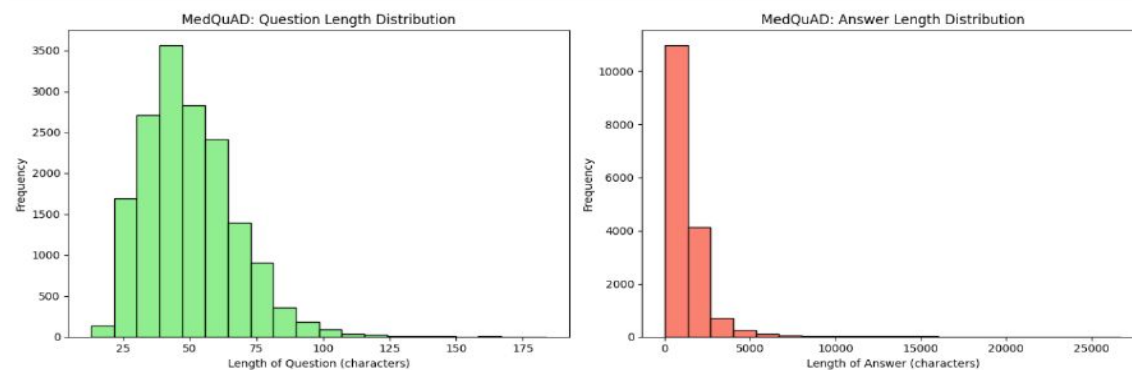


Visualizations

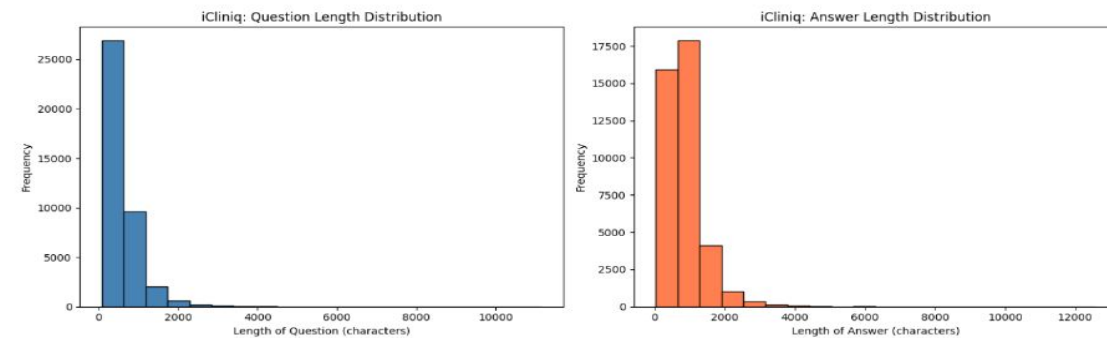




Distribution of text lengths across the Question and Answer columns



Distribution of text lengths across the Question and Answer columns



Model Selection

Selected open-source LLMs optimized for chronic disease management, ensuring accuracy, scalability, and patient engagement.

Selected Models:

- **MedLlama3:** General-purpose Q&A and patient education
- **BioMistral-7B:** Biomedical knowledge retrieval and clinical research
- **MedAlpaca:** Lightweight generative medical text and Q&A model,
- **Meditron:** Advanced medical reasoning and differential diagnosis

We have currently fine-tuned the BioMistral-7B and MedAlpaca models on healthcare-specific Q/A datasets, to enhance its performance in biomedical knowledge retrieval and clinical research queries.

Base-Model code for BioMistral-7B

Loading BioMistral model

```
# Block 4: Load BioMistral 7B Model
def load_model_and_tokenizer(model_name="BioMistral/BioMistral-7B"):
    """Load BioMistral 7B model and tokenizer with quantization."""
    tokenizer = AutoTokenizer.from_pretrained(model_name)
    if tokenizer.pad_token is None:
        tokenizer.pad_token = tokenizer.eos_token
    bnb_config = BitsAndBytesConfig(
        load_in_4bit=True,
        bnb_4bit_quant_type="nf4",
        bnb_4bit_compute_dtype=torch.float16,
        bnb_4bit_use_double_quant=True
    )
    model = AutoModelForCausalLM.from_pretrained(
        model_name,
        quantization_config=bnb_config,
        device_map="auto",
        torch_dtype=torch.float16,
        trust_remote_code=True
    )
    model.config.pad_token_id = tokenizer.eos_token_id
    model.eval()
    return model, tokenizer
```

```
model, tokenizer = load_model_and_tokenizer()
print(f"BioMistral-7B model and tokenizer loaded successfully!")
```

```
/opt/conda/lib/python3.10/site-packages/torch/_utils.py:831: UserWarning: Typ
e will be the only storage class. This should only matter to you if you are u
ped_storage() instead of tensor.storage()
    return self.fget.__get__(instance, owner)()
BioMistral-7B model and tokenizer loaded successfully!
```

Fine-Tuned Model

code for

BioMistral-7B

```
[12]: # Load fine-tuned model and tokenizer
def load_fine_tuned_model(model_name="BioMistral/BioMistral-7B", checkpoint_dir="biomistral_finetuned"):
    """Load the fine-tuned QLoRA model and tokenizer."""
    try:
        tokenizer = AutoTokenizer.from_pretrained(model_name)
        if tokenizer.pad_token is None:
            tokenizer.pad_token = tokenizer.eos_token

        bnb_config = BitsAndBytesConfig(
            load_in_4bit=True,
            bnb_4bit_quant_type="nf4",
            bnb_4bit_compute_dtype=torch.float16,
            bnb_4bit_use_double_quant=False,
            llm_int8_enable_fp32_cpu_offload=True,
            llm_int8_skip_modules=["lm_head"]
        )

        base_model = AutoModelForCausalLM.from_pretrained(
            model_name,
            quantization_config=bnb_config,
            device_map="auto",
            torch_dtype=torch.float16,
            trust_remote_code=True,
            offload_folder="offload",
            low_cpu_mem_usage=True,
            offload_state_dict=True
        )
        base_model.config.pad_token_id = tokenizer.eos_token_id

        if os.path.exists(checkpoint_dir):
            if os.path.exists(os.path.join(checkpoint_dir, "adapter_model.bin")):
                print(f"Loading fine-tuned model from {checkpoint_dir}")
                model = PeftModel.from_pretrained(base_model, checkpoint_dir, is_trainable=False)
            else:
                checkpoints = [d for d in os.listdir(checkpoint_dir) if d.startswith("checkpoint-")]
                if checkpoints:
                    latest_checkpoint = max(checkpoints, key=lambda x: int(x.split("-")[1]))
                    checkpoint_path = os.path.join(checkpoint_dir, latest_checkpoint)
                    print(f"Loading fine-tuned model from checkpoint: {checkpoint_path}")
                    model = PeftModel.from_pretrained(base_model, checkpoint_path, is_trainable=False)
                else:
                    raise ValueError(f"No checkpoints or final model found in {checkpoint_dir}")
            else:
                raise ValueError(f"Checkpoint directory {checkpoint_dir} does not exist")

        return model, tokenizer
    except Exception as e:
        print(f"Error loading fine-tuned model: {e}")
        return None, None

model, tokenizer = load_fine_tuned_model()
if model is None or tokenizer is None:
    raise ValueError("Failed to load fine-tuned model or tokenizer")
```

```
/opt/conda/lib/python3.10/site-packages/torch/_utils.py:831: UserWarning: TypedStorage is deprecated. It will be removed in the future and
UntypedStorage will be the only storage class. This should only matter to you if you are using storages directly. To access UntypedStorage
directly, use tensor.untyped_storage() instead of tensor.storage()
  return self._get_(instance, owner)()
Loading fine-tuned model from checkpoint: biomistral_finetuned/checkpoint-33573
```

Comparison of BioMistral Baseline and fine tuned

Metric	Baseline	Fine-Tuned
BERTScore Precision	0.8	0.8
BERTScore Recall	0.86	0.89
BERTScore F1	0.83	0.84
FCS (Factual Consistency Score)	0.3859	0.53
Hybrid BERT-BLEU	0.58	0.59
LLM Judge Score	0.52	0.84
GEval Score	0.44	0.75
METEOR	0.16	0.24
ROUGE-L	0.08	0.08
Entity F1	0.2	0.1
MCR (Medical Concept Recall)	0.2	0.2
BLEU	0.01	0.02

Comparison of BioMistral and MedAlpaca

Metric	BioMistral	MedAlpaca
BERTScore Precision	0.8	0.5
BERTScore Recall	0.89	0.52
BERTScore F1	0.84	0.5
FCS (Factual Consistency Score)	0.53	0.26
Hybrid BERT-BLEU	0.59	0.31
LLM Judge Score	0.84	0.42
GEval Score	0.75	0.31
METEOR	0.24	0.1
ROUGE-L	0.08	0.05
Entity F1	0.1	0.1
MCR (Medical Concept Recall)	0.2	0.1
BLEU	0.02	0.01

Thank You